

CSE220
Online on Convolution

Subsections:
Total Marks: 10
Total Time: 30 Minutes

Two cascaded LTI systems have the equivalent impulse response obtained by convolving their individual impulse responses. Consider

$$h_1[n] = \delta[n] - \delta[n-1] = [1, -1] \quad (\text{first difference}), \quad h_2[n] = u[n] \quad (\text{running-sum accumulator}).$$

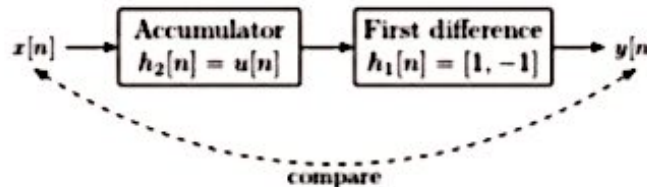
The accumulator has zero initial condition and computes

$$v[n] = \sum_{k=-\infty}^n x[k].$$

Since

$$(h_1 * h_2)[n] = u[n] - u[n-1] = \delta[n],$$

the cascade should act as the identity system. Instead of explicitly constructing $h_1 * h_2$, you will verify this using the non-impulse input signal given below and compare the final output with the original input.



In this online, reuse your **DiscreteSignal** and **LTISystem** classes from the offline. Represent both systems by their impulse responses and connect them using repeated calls to **LTISystem.output**.

Given

$$\begin{aligned} x[n] : \quad n &= -2 \dots 8, & \text{values} &= [2, -1, 3, 1, -2, 0, 0, 0, 0, 0], \\ h_1[n] : \quad n &= 0 \dots 1, & \text{values} &= [1, -1], \\ h_2[n] : \quad h_2[n] &= 1 \text{ for } n \geq 0. \end{aligned}$$

Finite-window note. The infinite sequence $u[n]$ cannot be stored. In the template, store $h_2[n] = 1$ for $n = 0 \dots 10$. This is enough for every sample of the given input to affect the complete graded observation window $n = -2 \dots 8$. Compare signals only on that window.

Tasks

- Implement `cascade(first_system, second_system, input_signal)`. Apply `first_system.output` first, then `second_system.output`, and return both the intermediate and final outputs.
- Implement `max_absolute_difference_in_range(first, second, start, end)` over the inclusive range `start...end`.
- Construct $x[n]$, $h_1[n]$, and the stored window of $h_2[n]$ using **DiscreteSignal**; then create the two **LTISystem** objects.
- Apply $x[n]$ through **Accumulator** → **First difference**. Print the input samples, accumulator-response samples, and first-difference-response samples for $n = -2 \dots 8$.
- Using the plotting helper provided in the template, plot $x[n]$, the accumulator response $v[n]$, and the first-difference response $y[n]$ over $n = -2 \dots 8$ as discrete-time stem plots.
- Compute and print the maximum absolute difference between the first-difference response and $x[n]$ over $n = -2 \dots 8$.
- In a print statement, conclude that the cascade output equals $x[n]$, so the cascade is an identity system and the differentiator and accumulator are inverse systems under zero initial conditions.

Restriction

Do not use `numpy.convolve`, `scipy.signal`, or any built-in convolution routine.

Submission

Rename your Python file as 2305xxx.py and submit it in Moodle.

can you explain the question?
and how $h_1 * h_2[n] = u[n] - u[n-1]$?

Sure. This question is actually testing **how two LTI systems cancel each other**. Let me explain the whole problem first, then I'll prove why

$$(h_1 * h_2)[n] = u[n] - u[n-1] = \delta[n].$$

What is the question asking?

The assignment gives you **two LTI systems connected in series (cascade)**.

Input $x[n]$

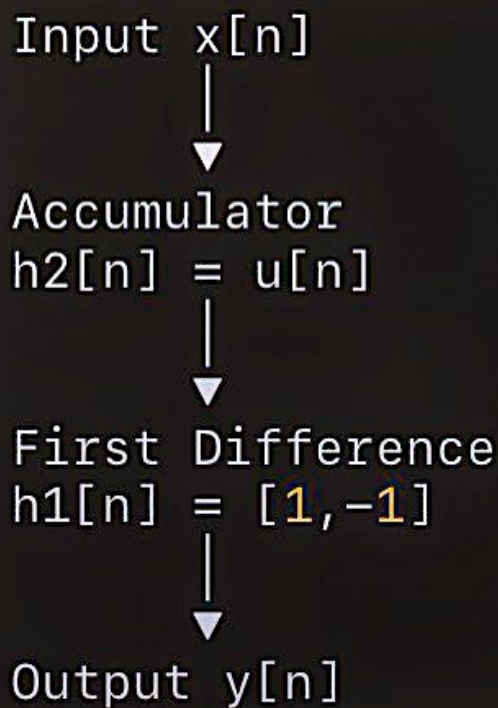


Accumulator
 $h_2[n] = u[n]$



First Difference
 $h_1[n] = [1, -1]$





Instead of sending the signal through one system, it passes through **both**.

For cascaded LTI systems,

$$h_{eq}[n] = h_1[n] * h_2[n]$$

where h_{eq} is the equivalent impulse response.

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The assignment **already** tells you

$$(h_1 * h_2)[n] = \delta[n]$$

and asks you to **verify it using an ordinary input signal**, not by directly convolving the impulse responses.

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So you'll

1. send $x[n]$ through the accumulator,
2. then send that output through the differentiator,
3. compare the final result with the original input.

If they are identical,

$$y[n] = x[n]$$

then the cascade acts as an **identity system**.

What does each system do?

System 1

Impulse response

$$h_1[n] = \delta[n] - \delta[n - 1]$$

which is

$$\begin{array}{rcl} n : & 0 & 1 \\ & 1 & -1 \end{array}$$

This is called the **first difference**.

Convolving any signal with this gives

$$y[n] = x[n] - x[n - 1]$$

It computes the difference between adjacent samples.

Example

Input

2 4 7 8

Output

2 2 3 1

because

2
4-2=2
7-4=3
8-7=1

System 2

Impulse response

$$h_2[n] = u[n]$$

where

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

This is an accumulator

Its output is

$$v[n] = \sum_{k=-\infty}^n x[k]$$

meaning

Output = running sum

Example

Input

2 4 7 8

Accumulator output

2
6
13
21

Why do they cancel?

Think of real numbers.

Suppose

Input

2
4
7
8

Accumulator gives

2
6
13
21

Now take differences

2
 $6 - 2 = 4$
 $13 - 6 = 7$
 $21 - 13 = 8$

The original numbers come back.

That is exactly why they are inverse systems.

Now the proof

We want

$$h_1 * h_2$$

Since

$$h_1[n] = \delta[n] - \delta[n - 1]$$

we use linearity:

$$(\delta[n] - \delta[n - 1]) * u[n]$$

=

$$\delta[n] * u[n] - \delta[n - 1] * u[n].$$

First term

One important convolution property is

$$\delta[n] * x[n] = x[n].$$

Therefore

$$\delta[n] * u[n] = u[n].$$

Second term

Another property is

$$\delta[n - k] * x[n] = x[n - k].$$

Here

$$k = 1.$$

Therefore

$$\delta[n - 1] * u[n] = u[n - 1].$$

So

$$h_1 * h_2 = u[n] - u[n - 1].$$

Why is

$$u[n] - u[n - 1] = \delta[n]?$$

Let's write both sequences.

Step function

$u[n]$

$n : -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3$

$0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1$



Shift it right



$u[n-1]$

$n : -2 \ -1 \ 0 \ 1 \ 2 \ 3$

0 0 0 1 1 1

Subtract them



$u[n]-u[n-1]$

$n : -2 \ -1 \ 0 \ 1 \ 2 \ 3$

0 0 1 0 0 0

But



$\delta[n]$

$n : -2 \ -1 \ 0 \ 1 \ 2 \ 3$

0 0 1 0 0 0



They are identical.

Hence

$$u[n] - u[n - 1] = \delta[n].$$

Therefore

$$h_1 * h_2 = \delta[n].$$

Since an impulse response of $\delta[n]$ is the identity system,

$$y[n] = x[n] * \delta[n] = x[n].$$

What your code will actually do

Your assignment explicitly says **do not compute $h_1 * h_2$ directly**. Instead, you should verify the result by cascading the systems.

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So your program will perform:

Original signal



Accumulator (running sum)



Intermediate signal $v[n]$



First Difference



Final output $y[n]$

Finally, compare

Original $x[n]$

with

Final $y[n]$

but how the convolution of h_2 and x will give v ?

Excellent question. This is actually the **core concept of LTI systems**. Let's derive it from the convolution formula instead of just accepting it.

Step 1: The output of an LTI system

For any LTI system,

$$y[n] = x[n] * h[n]$$

which means

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k].$$

Now substitute

$$h_2[n] = u[n].$$

So

$$v[n] = x[n] * u[n] = \sum_{k=-\infty}^{\infty} x[k] u[n-k].$$

Notice I'm calling the output $v[n]$ because the assignment calls the accumulator output

$v[n]$. [spec.pdf](#)

Step 2: What is $u[n-k]$?

Remember,

$$u[m] = \begin{cases} 1, & m \geq 0 \\ 0, & m < 0 \end{cases}$$

Replace m with $n-k$.

So

$$u[n-k] = \begin{cases} 1, & n-k \geq 0 \\ 0, & n-k < 0 \end{cases}$$

Rearrange the inequality:

$$n-k \geq 0$$

means



means

$$k \leq n.$$

Therefore,

$$u[n - k] = \begin{cases} 1, & k \leq n \\ 0, & k > n \end{cases}$$

This is the key observation.

Step 3: Substitute into the summation

We had

$$v[n] = \sum_{k=-\infty}^{\infty} x[k]u[n - k].$$

But for every $k > n$,

$$u[n - k] = 0.$$

Those terms disappear.

So only the terms where

$$k \leq n$$

remain.

Therefore

$$v[n] = \sum_{k=-\infty}^n x[k].$$

This is exactly the running sum (accumulator).

Why is it called an accumulator?

Suppose

$$x[n] = [2, -1, 3, 1].$$

Let's compute the convolution.

At $n = 0$

$$v[0] = \sum_{k=-\infty}^0 x[k].$$

Only



Only

$$x[0] = 2$$

exists.

So

$$v[0] = 2.$$

At $n = 1$

Now

$$v[1] = x[0] + x[1].$$

$$= 2 + (-1) = 1.$$

At $n = 2$

$$v[2] = 2 + (-1) + 3 = 4.$$

At $n = 3$

$$v[3] = 2 + (-1) + 3 + 1 = 5.$$

So the output becomes

n	$x[n]$	$v[n]$
0	2	2
1	-1	1
2	3	4
3	1	5

Each output is the **sum of all previous inputs**, which is exactly what an accumulator does.

A visual explanation

Suppose

$$x[n] = [2, -1, 3, 1].$$

For $n = 2$,



the convolution formula is

$$v[2] = \sum_k x[k]u[2 - k].$$

Let's evaluate $u[2 - k]$:

k	$u[2 - k]$
-----	------------

-2	1
----	---

-1	1
----	---

0	1
---	---

1	1
---	---

2	1
---	---

3	0
---	---

4	0
---	---

Notice that after $k = 2$, everything becomes 0, so those terms contribute nothing.

Therefore,

Therefore,

$$v[2] = x[-2] + x[-1] + x[0] + x[1] + x[2].$$

This is why convolution with the unit step **adds up all past samples**.

The intuition

Think of $u[n]$ as a **gate**:

- It lets all **past and current** samples pass (value 1).
- It blocks all **future** samples (value 0).

So during convolution, the unit step acts like a window that collects every sample up to the current time n , producing the cumulative (running) sum. That's why the assignment states that convolving with $h_2[n] = u[n]$ gives the accumulator output $v[n]$.

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then how the convolution of h_1 and v will be x ?

Exactly! This is the second half of the proof. Let's derive it from the convolution formula instead of memorizing it.

We already know from the assignment that

$$v[n] = \sum_{k=-\infty}^n x[k].$$

Now we send $v[n]$ into the first-difference system whose impulse response is

$$h_1[n] = \delta[n] - \delta[n - 1].$$

The output is

$$y[n] = v[n] * h_1[n].$$

Method 1: Using convolution properties (easiest)

Substitute $h_1[n]$:

$$y[n] = v[n] * (\delta[n] - \delta[n - 1]).$$

Using linearity,

$$y[n] = v[n] * \delta[n] - v[n] * \delta[n - 1].$$

Now use the delta properties:

$$x[n] * \delta[n] = x[n]$$

and

$$x[n] * \delta[n - 1] = x[n - 1].$$

Therefore,

$$y[n] = v[n] - v[n - 1].$$

So the differentiator simply computes the difference between consecutive accumulator outputs.

Now substitute the expression for $v[n]$:

$$v[n] = \sum_{k=-\infty}^n x[k]$$

and

$$v[n-1] = \sum_{k=-\infty}^{n-1} x[k].$$

Subtract them:

$$y[n] = \sum_{k=-\infty}^n x[k] - \sum_{k=-\infty}^{n-1} x[k].$$

Notice that almost every term cancels.

Write them out:

$$\begin{aligned} v[n] &= x[-\infty] + \dots + x[n-2] + x[n-1] + x[n] \\ v[n-1] &= x[-\infty] + \dots + x[n-2] + x[n-1]. \end{aligned}$$

Subtracting gives

$$y[n] = x[n].$$

Everything before $x[n]$ cancels!

Numerical example

Suppose

$$x[n] = [2, -1, 3, 1].$$

Step 1: Accumulator

n	x[n]	v[n]
0	2	2
1	-1	1
2	3	4
3	1	5

because

$$\begin{aligned} &2 \\ &2 + (-1) = 1 \\ &1 + 3 = 4 \\ &4 + 1 = 5 \end{aligned}$$



Step 2: First Difference

Now compute

$$y[n] = v[n] - v[n-1].$$

n	v[n]	v[n-1]	y[n]
0	2	0*	2
1	1	2	-1
2	4	1	3
3	5	4	1

(* The assignment assumes zero initial conditions, so $v[-1] = 0$.)

Now compare:

Original x[n]	Final y[n]
2	2
-1	-1

Original $x[n]$

Final $y[n]$

2

2

-1

-1

3

3

1

1

They are exactly the same.

Why are they called inverse systems?

Think of the accumulator as **adding**
everything up:

x
|
▼
2
1
4
5



Then the differentiator asks:

"How much did the running total increase since the previous sample?"


$$\begin{aligned} 2 \\ 1-2 &= -1 \\ 4-1 &= 3 \\ 5-4 &= 1 \end{aligned}$$

The increase at each step is precisely the original input sample.

The whole process



Original signal $x[n]$



Accumulator
 $v[n] = \sum x[k]$

The whole process



Original signal $x[n]$



Accumulator
 $v[n] = \sum x[k]$



First Difference
 $y[n] = v[n] - v[n-1]$



$x[n]$

So mathematically,

$$v[n] = \sum_{k=-\infty}^n x[k],$$

$$[4pt] y[n] = v[n] - v[n-1],$$

$$[4pt] = \left(\sum_{k=-\infty}^n x[k] \right) - \left(\sum_{k=-\infty}^{n-1} x[k] \right),$$

$$[4pt] = x[n].$$

This is exactly what your online assignment wants you to verify by simulation instead of directly computing $h_1 * h_2$.

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